

# Generation Interconnection System Impact Study Report

## For Queue Project Q21-1964

Hemet 138 kV  
December 29, 2022

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1964. The Interconnection Customer proposes a 105.4 MW solar photovoltaic facility interconnecting at the Hemet 138 kV substation in the Pennsylvania area. The request was received on December 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 680 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,797,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                            | Contingency                         | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------------|-------------------------------------|-------------|--------------|----------|
| Dormont-Tillery 138 KV                        | Hoyt-Collier 138 KV Line            | 134.8       | 181          | 6.0      |
| Causeway-Dormont 138 KV                       | Colier-Hoyt 138 KV Line             | 124.7       | 211          | 6.0      |
| Causeway-U S Alliance Coosa Pines 138 KV line | Hoyt-Collier 138 KV                 | 122.4       | 238          | 6.0      |
| U S Alliance Coosa Pines-Woodville 138 KV     | Collier-Hoyt Substation 138 KV line | 122.4       | 246          | 6.0      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                  | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Dormont-Tillery 138 KV                    | Rebuild line with higher-capacity conductor | \$1,302,000   | 100% (\$1,302,000)    |
| Causeway-Dormont 138 KV                   | Rebuild line with higher-capacity conductor | \$742,000     | 100% (\$742,000)      |
| Causeway-U S Alliance Coosa Pines 138 KV  | Rebuild line with higher-capacity conductor | \$1,215,000   | 100% (\$1,215,000)    |
| U S Alliance Coosa Pines-Woodville 138 KV | Rebuild line with higher-capacity conductor | \$3,538,000   | 100% (\$3,538,000)    |

| Facility | Required Upgrade           | Cost Estimate | Allocation to Project |
|----------|----------------------------|---------------|-----------------------|
|          | Total allocated to project |               | \$6,797,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.